

Comprehensive Quiz Exam (27 Feb 2016)

Name::

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1. Which of the following **is** produced by the polymerization of carbohydrate monomers?
 - a. Lipids
 - b. Polysaccharides
 - c. Proteins
 - d. Nucleic acids
2. CO₂ is carried in blood mainly as
 - a. Attached to hemoglobin
 - b. Bicarbonate ion (HCO₃⁻)
 - c. Dissolved CO₂ in blood
 - d. CO
3. Which of the following is an amino acid?
 - a. Alanine
 - b. Phenylalanine
 - c. Glutamic acid
 - d. Proline
4. Which statement **is not** correct regarding the use of sugars by the cells?
 - a. Sugars are used by some cells for formation of cell wall
 - b. Sugars are used by cells for energy production
 - c. Sugars are used by cells to catalyze biochemical reactions
 - d. Sugars are used for cell-cell communication
5. In liquid water every H₂O is hydrogen bonded to how many neighboring water molecules?
 - a. 3.4
 - b. 2.4
 - c. 1.4
 - d. 4.4
6. Which pair of amino acid is polar and acidic?
 - a. Val-Glu
 - b. Glu-Cys
 - c. Glu- Ser
 - d. Trp-Glu
7. Which amino acid has imidazole group?
 - a. Arg
 - b. His
 - c. Lys
 - d. Tyr
8. Which group of amino acids is aromatic?
 - a. Phe, Arg, Lys
 - b. Phe, Tyr, Pro
 - c. Phe, Tyr, Trp
 - d. Met, Trp, His
9. Which amino acid has sulfur in its structure?
 - a. Phe, Tyr

- b. Lys, His
 - c. Cys, Met
 - d. Gly, Ala
10. The pK_a's of lysine are 2.18, 8.95, and 10.79. The correct isoelectric point of lysine is:
- a. 9.87
 - b. 2.18
 - c. 8.955
 - d. 10.79
11. At pH = 1, the R group of acidic and basic amino acids are charged.
- a. True
 - b. False, only basic are charged
 - c. False, neither acidic nor basic are charged
 - d. False, only acidic are charged
12. Tertiary structure is stabilized by which of the following interactions?
- a. hydrophobic interactions
 - b. hydrogen bonds
 - c. ionic bonds
 - d. disulfide bonds
 - e. all of the above
13. In proteins, elements of secondary structure combine to form a(n)
- a. Domain
 - b. Motif
 - c. Alpha helix
 - d. Beta sheet
14. Beta pleated sheet is
- a. Primary structure
 - b. Secondary structure
 - c. Tertiary structure
 - d. Quaternary structure
15. The structure formed by covalent bonds between adjacent amino acid residues in a polypeptide is called
- a. Primary structure
 - b. Secondary structure
 - c. Tertiary structure
 - d. Quaternary structure
16. All twenty amino acids are chiral.
- a. False
 - b. True
 - c. Not given
17. Draw the structure of Aspartate?
18. Write the structure of lysine as it would be expected at its isoelectric point.
19. Write the dissociation of acetic acid and the **Henderson-Hasselbach Equation** for it.
20. Write the structure of tryptophan as it would be expected at pH 7.